

Sienna Talles

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University junior seeking opportunities in quantitative research and data science. Experienced in statistical modeling, machine learning, and data analysis using Python, R, and SQL, with a strong interest in applying quantitative methods to business problems.

EDUCATION

Washington University in St. Louis

Bachelor of Science in Data Science, Minors in Mathematics and Business Analytics
Dean's List, GPA 3.96/4.0

St. Louis, MO

Expected Graduation: May 2028

The London School of Economics and Political Science (LSE) Summer School

Introduction to Statistics: Understanding the World through Data – Participant

London, England

June-July 2025

Examined how data is applied to solve real-world issues. Analyzed datasets through visualizations, numerical summaries, and modeling.

SKILLS AND PROGRAMMING

Python (libraries: Matplotlib, NumPy, Pandas, Scikit-learn, Seaborn), R, SQL, Java, HTML, JavaScript, CSS, Excel

WORK EXPERIENCE

Ipsos

New York, NY

Market Research Intern (Quantitative Analytics) - Innovation Team

June 2026-Present

- Assisted with survey design, tested live survey links for functionality and logic, and cleaned the resulting data, maintaining data integrity across concept and product studies for clients in financial services, retail, and technology.
- Applied statistical significance testing and analyzed output from MaxDiff and TURF simulators alongside quantitative and qualitative survey results, identifying key findings that shaped actionable recommendations delivered in client reports.

Washington University in St. Louis

St. Louis, MO

Personality Measurement and Development Lab - Undergraduate Research Assistant

January-August 2026

- Built a Python web scraping framework extracting 125,000+ variables across 600+ datasets from the German Socio-Economic Panel, handling multithreaded requests, retry logic, and API pagination.
- Restructured data into JSON/CSV pipelines to support variable prioritization for a model predicting personality-based life outcomes.

CSE 1301: Introduction to Computer Science Teaching Assistant

August 2025-Present

- Guided students through Java programming fundamentals and debugging during weekly studio sessions and office hours, breaking down core computer science concepts into clear, approachable explanations.
- Graded assignments and assessments for accuracy and logic, providing feedback to reinforce correct problem-solving approaches.

PROJECTS

Optiver Realized Volatility Prediction Competition

September 2026

Independent Quantitative Researcher

- Competed in Optiver's Kaggle Code competition to generate short-term signals from order book and trade data of a fixed 10-minute window to predict the realized volatility of the next 10-minute window.
- Analyzed 429,000 windows across 112 stocks, using weighted average price to compute log returns and realized volatility.
- Engineered book, trade, and market-wide features, including realized volatility over smaller windows (past 5 and 2 minutes), bid-ask spread, trading activity, and average realized volatility across all stocks in the same window.
- Compared naive baseline, Linear Regression, and LightGBM models with time-window splits, reducing RMSPE from 0.341 to 0.249.

Earnings Announcement Effects on Options Implied Volatility

September 2026

Independent Quantitative Researcher

- Investigated whether options expiring shortly after a company's earnings announcement carry higher implied volatility than later-dated options on the same stock, using live option chains for 31 large-cap tickers across technology, financials, healthcare, energy, and consumer sectors (e.g., AAPL, JPM, UNH, XOM) via the yfinance API.
- Engineered a variance-decomposition metric to isolate the earnings effect from normal volatility term structure and validated with a one-sided t-test to conclude the effect was statistically significant ($p = 0.01$).
- Built an automated daily pipeline that tracks the same short-term option per ticker through its earnings announcement, observing implied volatility rising into each earnings date and dropping sharply afterward.

ASA DataFest - Analyzing Encounter Duration in the SVH System

April 2026

Best Use of Statistical Analysis Award

- Analyzed hospital encounter data in a 48-hour ASA DataFest competition to identify drivers of patient length of stay; pinpointed departments, seasons, and diagnoses associated with longer stays to inform resource allocation.
- Cleaned and explored seven healthcare datasets in Python, engineered patient length of stay from admission and discharge timestamps, and built Linear Regression and Random Forest models for prediction.

CAMPUS INVOLVEMENT

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- Alpha Phi, *Member*: Involved in philanthropy initiatives (e.g. heart health), and membership recruitment
- Data Science Network, *Member*: Attend meetings, learn about data science careers, knowledge sharing, networking
- Association for Women in Mathematics, *Member*: Attend meetings, knowledge sharing, networking